

CS & IT ENGINEERING



Data Structure & Programming

Arrays

Lecture No.- 01



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Topics to be Covered



- Data structure ?
- Classification
- Time Complexity and Space Complexity Introduction
- Arrays - Insertion, Deletion, Search

SYLLABUS

- Time Complexity, Space Complexity

Linear Data Structures

- Arrays
- Linked List
- Stack
- Queue
- Hash Table

Non-linear Data Structures

- Trees
- Graphs



PDS WEIGHTAGE



YEAR	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
MARKS	<u>15</u>	<u>11</u>	<u>11</u>	<u>8</u>	<u>12</u>	<u>12</u>	<u>10</u>	<u>14</u>	<u>12</u>	<u>18</u>

Avg Weightage : 12 + Marks

Reference Book: "Data Structures made Easy" by Narasimha Karumanchi



Topic : Arrays - 1



Data : Collection of raw facts.

Information : Processed data / filtered data

Data : Text (Numbers, letters, Symbols), Audio, Video, Images [multimedia]

Data Type vs Data Structure

Describes

- Nature / characteristics
- Memory Requirement
- Range of Values

Describes

- Organization (or) Management.



Topic : 1-D Arrays



Data Structure

- Organization (or) Management of data
- It means, How various operations are Performed using/on data? is defined.

- Operations like :

- Creation / construction
- Add / Insertion
- Modify
- Delete / Remove
- Search / Find
- Sorting
- Traverse / Access

- Data Structure can also be defined as representation of data.

Examples :



Lists



Table



Graph



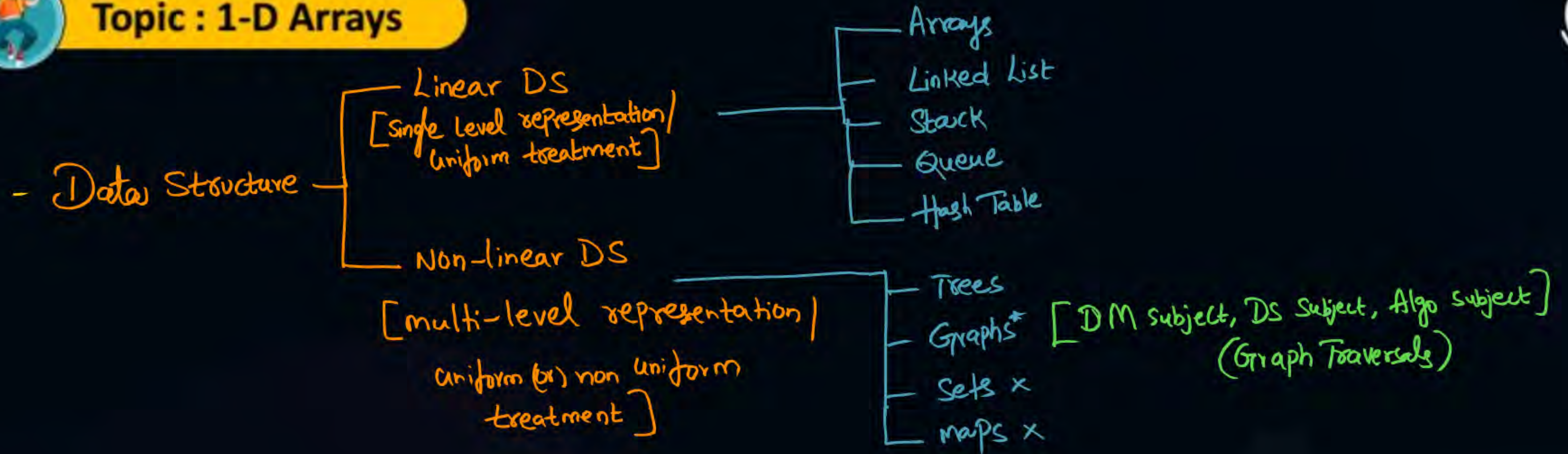
Pie-chart



sets/Venn diagram



Topic : 1-D Arrays



- Problem Solving → Program / Software →
- 1) Understand Problem, Constraints, Feasibility Study
 - 2) Algorithm : A simple step-by-step Procedure
 - 3) Flow Chart : Diagrammatic representation
 - 4) Pseudo Code : Syntax free version of Program
 - 5) Program / Software
- 1 : Many Relationship
- Problem : multiple ways to solve
[multiple Algorithms]



Topic : 1-D Arrays



Algorithm

- Efficiency of an algorithm is measured Using 2 Parameters

1) Time Complexity : The Execution (or) running time of an algorithm

2) Space Complexity : The memory requirement of data* also code in the algorithm.

- To measure Time and Space Complexity, An algorithm is analyzed at 2 phases:

1) Priori Analysis

[Before Prg/sw is ready]

- Approximated results

2) Posterior analysis [After Program/software is ready $\equiv$ Testing]

- Accurate results.

- Asymptotic Notations: Used to represent the measured T.C, S.C



Topic : 1-D Arrays



- Time Complexity & Space Complexity measured/observed in 3 different cases:

- Best case

- Average case

- Worst case

Omega Notation
 $\Omega()$

Theta Notation
 $\Theta()$

Big-oh Notation
 $O()$

- $O(1) < O(\log n) < O(n) < O(n \log n) < \underbrace{O(n^2) < O(n^3) \dots}_{\text{Polynomial Time}} < \underbrace{O(2^n)}_{\text{Exponential time}} < \underbrace{O(n^n)}_{\text{Highest}}$

Lowest Constant time Linear Time



A one dimensional array A has indices 1....75. Each element is a string and takes up three memory words. The array is stored at location 1120 decimal. The starting address of A[49] is _____



Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in a non-decreasing order. The distance of the array [2, 5, 3, 1, 4, 2, 6] is _____



Suppose there are 11 items in sorted order in an array. How many searches are required on the average, if binary search is employed and all searches are successful in finding the item?



Let A be an array of 31 numbers consisting of sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i that $A[i]$ is 1 by probing the minimum numbers of locations in A. The worst-case number of probes performed by an optimal algorithm is _____.



Topic : 1-D Arrays – ISRO 2013

H/W



Let $A(1:8, -5:5, -10:5)$ be a three dimensional array. How many elements are there in the array A?



An array A consists of n integers in locations $A[0], A[1] \dots A[n-1]$. It is required to shift the elements of the array cyclically to the left by k places, where $1 \leq k \leq (n-1)$. An incomplete algorithm for doing this in linear time, without using another array is given below. Complete the algorithm by filling in the blanks. Assume all the variables are suitably declared.

```
min = n; i = 0;
while ( _____ ) {
    temp = A[i]; j = i;
    while ( _____ ) {
        A[j] = _____
        j = (j + k) mod n ;
        If ( j < min ) then
            min = j;
    }
    A[(n + i - k) mod n] = _____
    i = _____
```

- A. $i > \text{min}; j \neq (n+i) \bmod n; A[j + k]; \text{temp}; i + 1;$
- B. $i < \text{min}; j \neq (n+i) \bmod n; A[j + k]; \text{temp}; i + 1;$
- C. $i > \text{min}; j \neq (n+i+k) \bmod n; A[(j + k)]; \text{temp}; i + 1;$
- D. $i < \text{min}; j \neq (n+i-k) \bmod n; A[(j + k) \bmod n]; \text{temp}; i + 1;$



Topic : 1-D Arrays



```
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while ( _____ ) {
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D. $i < \text{min}; j \neq (n+i-k) \bmod n; A[(j + k) \bmod n]; \text{temp}; i + 1;$



2 mins Summary



- DS
- Classification of DS
- T.C, S.C
- Notations

To be contd ...





THANK - YOU